



National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

Object Oriented Programming Lab Semester Project

Project Title: Chess Game in C++

Objective:

The goal of this project is to design and implement a Chess Game using the concepts of Object-Oriented Programming (OOP) in C++.

Students should apply OOP principles such as classes, objects, inheritance, polymorphism, composition, and encapsulation to create a working and well-structured program.

Project Description:

You are required to create a simple **two-player Chess Game** that can be played in the console (text-based). The game should allow two players to play against each other by taking turns and making valid moves according to the rules of chess.

Key Features to Include:

- **Chess Board Setup:** An 8×8 chessboard displayed using text or symbols, with proper arrangement of all chess pieces at the start.
 - **Chess Pieces:** Each piece (Pawn, Rook, Knight, Bishop, Queen, King) should be represented by a class. All pieces should inherit from a common base class, e.g., `Piece`.
 - **Valid Moves:** Each piece should follow its correct movement rules. The program should not allow invalid moves.
 - **Turns:** Players should play alternately (White first, then Black).
 - **Basic Game Rules:** Detect when a King is captured or when checkmate occurs. Optionally, implement stalemate or draw conditions.
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OOP Concepts to be Demonstrated:

- **Encapsulation:** Use classes and private data members.
 - **Inheritance:** Derive piece classes (e.g., Pawn, Rook) from a base Piece class.
 - **Polymorphism:** Use virtual functions for movement validation.
 - **Composition:** Include a Board class that contains Piece objects.
 - **Graphical UI:** Bonus Marks.
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Project Deliverables:

1. **Source Code.**
 2. **A short project report** (2–3 pages) including:
 - Introduction.
 - Class diagram.
 - Explanation of how OOP concepts were used.
 - Screenshots of output.
 3. **Demonstration:** Students must run the project and explain their code.
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Github Submission:

The semester project must be submitted through GitHub, where each group will maintain a repository containing their complete project work.

Each Group Must be using Github to

Group size: 2–3 students

- Repository: one GitHub repository per group
- Each member must:
 - create their own GitHub account
 - be added as a collaborator in the group repository
 - make commits from their own account
- A shared account is not allowed

Each member must make a minimum number of meaningful commits

Example rule:

- each member must make at least 3–5 meaningful commits
- commit messages like:
 1. Added Book class
 2. Implemented issue/return function
 3. Designed main menu
 4. Fixed file handling bug

If anyone is face any issues regarding Github submission they can contact their Lab Instructor.

